

Massimiliano Berardi

Based in The Hague, Netherlands | massimiliano.berardi93@gmail.com | github.com/Max593

Experience

Software Engineer, Syntho – Amsterdam

September 2023 – Present

- **Backend development:** Designed and implemented backend services in FastAPI and Django across core platform components, improving API integration and maintainability through structured service design and platform-level feature development.
- **Database test images:** Extended and optimized Docker-based test images for MySQL/MariaDB, PostgreSQL, Microsoft SQL Server, OracleDB, IBM DB2, Apache Hive, and Sybase ASE, broadening schema coverage while reducing setup time from minutes to seconds and shortening test pipeline execution through leaner images and more targeted test scenarios.
- **SQLAlchemy connectors:** Built and maintained custom SQLAlchemy connectors across multiple SQL and non-SQL backends, improving consistency and support for backend-specific access patterns by extending SQLAlchemy with decorator-based overrides for database-specific behavior.
- **Complex ORM queries:** Designed SQLAlchemy ORM queries and data access patterns for reporting, filtering, and large-scale operations, improving correctness and consistency across multiple database backends through performance-conscious query design.
- **Data processing workflows:** Improved internal data processing workflows through greater reliability and more efficient handling of data structures, by building and maintaining pipelines with NumPy, PyArrow, and Spark DataFrames.
- **Docker deployment:** Improved Docker-based application deployment across development, CI, and production-like environments by refining container configuration and startup logic to increase portability and consistency.
- **Schema design:** Contributed to database schema and storage refactors for application artifacts and logs, improving maintainability, portability, and handling of large generated outputs across environments.

Software Engineering Intern, Syntho – Amsterdam

August 2021 – September 2023

- **Subsetter:** Designed and implemented a SQL database subsetter that preserved referential integrity across related tables, enabling smaller production-like datasets for development and testing by processing tables in topological dependency order.
- **PII scanner:** Contributed in a team of three to a PII detection tool that increased personal-data detection coverage by almost 30% in internal testing across structured and unstructured sources by combining Presidio, SpaCy, and custom regex patterns.
- **Test environment:** Created Docker-based test images for MySQL, PostgreSQL, and Microsoft SQL Server, improving consistency and repeatability of unit and integration testing by standardizing production-like environments across development and CI.

Skills

Programming: Python, SQL, Bash, Java, C, C++, TypeScript

Frameworks: FastAPI, Django, Ray, Pytest, Unittest

Libraries: SQLAlchemy, NumPy, Pandas, PyArrow, PySpark

Databases: PostgreSQL, MySQL/MariaDB, Microsoft SQL Server, OracleDB, IBM DB2, Apache Hive, Sybase ASE

Tools: Linux, Git, Docker, AWS, Poetry, Maven, SSH

Education

Rijksuniversiteit Groningen – BSc in Computing Science

September 2019 – July 2022

- **Teaching Assistant** Advanced Object Oriented Programming

April 2022 – June 2022

- **Teaching Assistant** Object Oriented Programming

September 2021 – December 2021

Thesis published as a peer-reviewed article:

M. Berardi, M. Penca, R. D. Boza, D. Feitosa, V. Andrikopoulos, “Mining for cost awareness in the infrastructure as code artifacts of cloud-based applications: An exploratory study”, Journal of Systems and Software, Volume 215

doi.org/10.1016/j.jss.2024.112112